

SAFDAR AYUB

AI Engineer | Full Stack Developer | Cloud-Native Kubernetes

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PROFESSIONAL SUMMARY

Results-oriented AI Engineer and Full Stack Developer with 23+ years of distinguished service in the Pakistan Air Force. Specializing in agentic AI systems, autonomous agent orchestration, and cloud-native technologies. Hands-on experience building production AI-powered applications using OpenAI Agents SDK, Claude Code, MCP Protocol, and modern full-stack frameworks. Military leadership background translating to exceptional project management and quality assurance capabilities.

Agentic AI | MCP Servers | Full Stack Development | Cloud-Native Kubernetes | Spec-Driven Development | Docker | E-Commerce | CRM & Dashboard Systems | Team Leadership

TECHNICAL SKILLS

AI & Agentic Systems:	OpenAI Agents SDK, Claude Code (Opus), MCP SDK, FastMCP, Prompt Engineering, RAG Systems, Gemini TTS
ML & Data Science:	scikit-learn (TF-IDF, KMeans, LogisticRegression, LinearSVC, RandomForest), 5-fold CV, Groq API (Llama 3.3 70B), aiohttp, gspread, Alembic
Programming Languages:	Python, TypeScript, JavaScript
Frontend:	Next.js, React.js, Remotion, Tailwind CSS, ShadCN UI, Recharts, Framer Motion, Zustand
Backend & Databases:	FastAPI, Node.js, REST APIs, Prisma, SQLAlchemy, PostgreSQL, SQLite, Turso/LibSQL, Neon DB, Supabase, Qdrant Cloud, Upstash Redis
Payments & Media:	Stripe Checkout, Cloudinary, browser-image-compression
DevOps & Cloud:	Docker, Kubernetes, Helm Charts, OCI/OKE, NGINX Ingress, Dapr, Apache Kafka, GitHub Actions, Vercel
Validation & Auth:	Zod, React Hook Form, NextAuth.js v5, Clerk, Better Auth, JWT, Gmail OAuth2

KEY PROJECTS

Personal AI Employee — Autonomous Agent System (Hackathon Platinum Tier) | [GitHub](#)

- Built local-first AI agent monitoring Gmail, filesystem, and WhatsApp with priority-queue orchestration and human-in-the-loop approval gates
- Implemented 4 custom FastMCP servers (email, social media, Odoo ERP, documents) with circuit breaker patterns and correlation ID tracking
- Designed risk-based action routing (routine/sensitive/critical) with dry-run defaults and PM2 process management
- Integrated Obsidian vault pipeline with 12 Claude Code skills for automated task processing and weekly CEO briefing generation

Tech Stack: Python, FastMCP, Claude Code, Gmail OAuth2, Playwright, Odoo RPC, APScheduler, Obsidian, PM2

AI/ML Job Market Intelligence Pipeline | [GitHub](#)

- Scraped 97 AI/ML jobs from RemoteOK with aiohttp/asyncio; stored with SQLAlchemy ORM and Alembic migrations on PostgreSQL
- Clustered jobs using silhouette-optimized KMeans (k=8) with TF-IDF vectorization, surfacing meaningful segments: Research, Applied ML, MLOps, NLP, CV, and more
- Built 3-model seniority classifier (Logistic Regression, LinearSVC, Random Forest) with 5-fold cross-validation, auto-selecting top performer by accuracy + F1
- Generated natural-language market briefings via Groq API (Llama 3.3 70B) and pushed dated tab reports to Google Sheets — entire pipeline triggered by a single POST /run-pipeline endpoint

Tech Stack: Python, aiohttp, PostgreSQL, SQLAlchemy, Alembic, scikit-learn (TF-IDF, KMeans, LR, LinearSVC, RF), Groq API (Llama 3.3 70B), FastAPI, gspread, Docker, pytest

AHF Auto Parts — Production JDM E-Commerce Platform | [GitHub](#) | [Live](#)

- Built production-grade e-commerce platform for a Japan-based auto parts business — vehicle compatibility lookup (Make → Model → Year cascade), product catalog with search/filter/sort, and image gallery
- Implemented Stripe Checkout and manual bank transfer with 30-minute stock reservation (Upstash Redis) to prevent overselling during concurrent checkout sessions
- Delivered full admin dashboard with Cloudinary image upload, order management, customer list, vehicle/category CRUD, contact inbox, and Recharts analytics
- Architected SSG + ISR (60s) for storefront pages and SSR for auth-protected routes; Google OAuth + credentials via NextAuth.js v5; transactional emails via Resend

Tech Stack: Next.js 16, TypeScript, PostgreSQL/Neon, Prisma, NextAuth.js v5, Stripe, Cloudinary, Zustand, Upstash Redis, Resend, Vercel

Campaign Manager — Full-Stack CRM Dashboard | [GitHub](#) | [Live Demo](#)

- Built a full-stack CRM prototype with contacts CRUD, audience segmentation rule builder, multi-step campaign wizard, and analytics dashboard

- Implemented server-rendered dashboard with KPI cards and Recharts analytics — zero client-side data waterfalls
- Designed visual segment rule builder with AND logic, multiple operators, and live preview with 300ms debounce
- Engineered Prisma v7 dual adapter pattern (SQLite locally, Turso LibSQL in production) with Zod-validated API routes

Tech Stack: Next.js 16, TypeScript, Tailwind CSS, ShadCN UI, Prisma, Turso/LibSQL, Zod, Recharts

AI Video Generation Agent — End-to-End Content Automation | [GitHub](#) | [YouTube Demo](#)

- Built AI pipeline that generates complete motion graphics videos from text prompts: script to storyboard to scenes to narration to rendered video
- Integrated Claude Code (Opus) as orchestrator with Remotion/React rendering engine and Gemini TTS for synchronized voiceover
- Produced 4-minute 1080p explainer video with spring physics animations and 3D card transformations

Tech Stack: TypeScript, React, Remotion, Claude Code (Opus), Gemini TTS, Spring Physics, Lucide React

AI-Powered Todo Application (8-Phase Cloud-Native Evolution) | [GitHub](#) | [Live](#)

- Evolved application across 8 phases: Console App to Full-Stack to AI Chatbot to Kubernetes to Kafka Events to Dapr Microservices to OCI Cloud
- Integrated AI chatbot using Groq API for natural language task management with conversation history
- Deployed production system on OCI OKE with Helm charts, NGINX Ingress, Dapr sidecar injection, and free-tier optimization

Tech Stack: Next.js, FastAPI, SQLAlchemy, Groq API, Docker, Kubernetes, Helm 3, Dapr, Apache Kafka, OCI OKE, Neon PostgreSQL

Physical AI & Humanoid Robotics Interactive Textbook | [GitHub](#) | [Live](#)

- Built interactive textbook with RAG chatbot using OpenAI Agents SDK, FastAPI, Neon Postgres, and Qdrant Cloud
- Created 6-chapter curriculum covering ROS 2, Gazebo, NVIDIA Isaac, and VLA models with user authentication

Tech Stack: Docusaurus, React, FastAPI, OpenAI Agents SDK, MCP SDK, Neon DB, Qdrant, Better Auth, Vercel

PROFESSIONAL EXPERIENCE

Quality Control Inspector & Electronics Technician (Retired) | Pakistan Air Force

February 2000 – March 2023 (23 Years)

- Led quality control inspection in Air Logistics and engineering depots ensuring adherence to military-grade standards
- Specialized in electronics and communication equipment maintenance across multiple operational units
- Managed cross-functional teams, HR development, and protocol liaison with foreign personnel

EDUCATION & CERTIFICATIONS

Certified Agentic and Robotic AI Engineer | PIAIC, Air University Islamabad | Expected 2026

Master of Computer Science (CGPA: 3.11) | Virtual University of Pakistan | 2012

DAE Radio Technology (71%) | School of Electronics, PAF Korangi Creek | 2003

- Artificial Intelligence and Computing Program — Presidential Initiative for AI and Computing (2023)
- PCAP: Programming Essentials in Python — Cisco Networking Academy / OpenEDG Python Institute (2022)

ADDITIONAL

- Veteran: 23+ years of distinguished service in Pakistan Air Force
- Active Contributor: Building production AI agents and open-source projects on GitHub
- Operating Systems: Linux Ubuntu, Windows